

Trigonometry

ANSWER KEY



The unit circle and radian measure

- 1 Convert 210° to radians.

$$210 \times \frac{\pi}{180} = \frac{7\pi}{6}.$$

- 2 Convert $\frac{5\pi}{4}$ radians to degrees.

$$\frac{5\pi}{4} \times \frac{180}{\pi} = 225^\circ.$$

- 3 Find the exact value of $\sin(240^\circ)$.

240° is in Quadrant III with reference angle 60° , where sine is negative: $\sin(240^\circ) = -\frac{\sqrt{3}}{2}$.

Trigonometric identities

- 4 Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$.

$$\frac{\cos^2 \theta}{\cos \theta} = \cos \theta.$$

- 5 Simplify $\tan \theta \cdot \cos \theta$.

$$\frac{\sin \theta}{\cos \theta} \cdot \cos \theta = \sin \theta.$$

- 6 If $\sin \theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos \theta$ and $\tan \theta$.

$$\cos \theta = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5} \text{ (negative in Quadrant II). } \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}.$$

Double angle formulas

- 7 Given $\cos \theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

$$\sin \theta = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}. \sin 2\theta = 2\left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}. \cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{288}{169} = -\frac{119}{169}.$$

- 8 Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin \theta$ and $\cos \theta$ only.

$$\sin 4\theta = 2\sin 2\theta \cos 2\theta = 2(2\sin \theta \cos \theta)(1 - 2\sin^2 \theta) = 4\sin \theta \cos \theta (1 - 2\sin^2 \theta).$$

Solving trigonometric equations

- 9 Solve $2\sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

$$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ \text{ or } \theta = 150^\circ.$$

- 10 Solve $2\cos^2 \theta + \cos \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Factoring: $(2\cos \theta - 1)(\cos \theta + 1) = 0$. $\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ, 300^\circ$. $\cos \theta = -1 \Rightarrow \theta = 180^\circ$.
Solutions: $60^\circ, 180^\circ, 300^\circ$.

Graphs of trigonometric functions

- 11 The graph shows $y = 3\sin(2x)$. State the amplitude and period.

$$\text{Amplitude} = 3. \text{ Period} = \frac{2\pi}{2} = \pi.$$

- 12 The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

$$\text{Amplitude} = 4. \text{ Period} = \frac{2\pi}{3}. \text{ Writing } 3x - \pi = 3\left(x - \frac{\pi}{3}\right), \text{ the phase shift is } \frac{\pi}{3} \text{ to the right.}$$

The Law of Sines

- 13 In triangle ABC , $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b , to the nearest hundredth.

$$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{10\sin 65^\circ}{\sin 40^\circ} \approx \frac{9.063}{0.643} \approx 14.10.$$

- 14 In triangle ABC , $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

$$h = b\sin A = 10\sin 35^\circ \approx 5.74. \text{ Since } h < a < b, \text{ two triangles are possible.}$$
$$\sin B = \frac{b\sin A}{a} \approx 0.717 \Rightarrow B \approx 45.8^\circ \text{ or } B \approx 180^\circ - 45.8^\circ = 134.2^\circ. \text{ Both are valid since } A + B < 180^\circ \text{ in each case.}$$

The Law of Cosines and the sine area formula

- 15 In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c , to the nearest hundredth, and the area of the triangle, to the nearest tenth.

$$c^2 = 11^2 + 14^2 - 2(11)(14)\cos 55^\circ \approx 317 - 176.66 = 140.34, \text{ so } c \approx 11.85. \text{ Area}$$
$$= \frac{1}{2}(11)(14)\sin 55^\circ = 77\sin 55^\circ \approx 63.1.$$

- 16 State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a , b , c and included angle C .

$$\text{Law of Cosines: } c^2 = a^2 + b^2 - 2ab\cos C. \text{ Area} = \frac{1}{2}ab\sin C.$$

Inverse trigonometric functions

- 17 Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.

$\frac{\pi}{4}$.

- 18 Evaluate $\arccos(-1)$ exactly, in radians.

π .

- 19 Find $\theta = \arctan(1)$ in degrees (principal value).

45° .

A well-designed word problem: locating a ship from two lighthouses

- 20 This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.

(a) $\angle ACB = 180^\circ - 72^\circ - 58^\circ = 50^\circ$ (b) AC is opposite $\angle B$: $AC = \frac{AB \sin B}{\sin C} = \frac{15 \sin 58^\circ}{\sin 50^\circ} \approx 16.6$ km. (c) BC is opposite $\angle A$: $BC = \frac{AB \sin A}{\sin C} = \frac{15 \sin 72^\circ}{\sin 50^\circ} \approx 18.6$ km.

Multiple choice — easy

- 21 Convert 135° to radians.

a. $\frac{3\pi}{5}$

b. $\frac{3\pi}{2}$

c. $\frac{3\pi}{4}$ ✓

d. π

$135 \times \frac{\pi}{180} = \frac{3\pi}{4}$.

22 Convert 60° to radians.

a. $\frac{2\pi}{3}$

b. π

c. $\frac{\pi}{4}$

d. $\frac{\pi}{3}$ ✓

$$60 \times \frac{\pi}{180} = \frac{\pi}{3}.$$

23 Convert 345° to radians.

a. 2π

b. $\frac{23\pi}{13}$

c. $\frac{23\pi}{12}$ ✓

d. $\frac{23\pi}{6}$

$$345 \times \frac{\pi}{180} = \frac{23\pi}{12}.$$

24 Convert 105° to radians.

a. $\frac{2\pi}{3}$

b. $\frac{7\pi}{13}$

c. $\frac{7\pi}{6}$

d. $\frac{7\pi}{12}$ ✓

$$105 \times \frac{\pi}{180} = \frac{7\pi}{12}.$$

25 Convert 90° to radians.

a. π

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{2}$ ✓

$$90 \times \frac{\pi}{180} = \frac{\pi}{2}.$$

26 Convert 30° to radians.

a. $\frac{\pi}{3}$

b. $\frac{\pi}{7}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$ ✓

$$30 \times \frac{\pi}{180} = \frac{\pi}{6}.$$

27 Convert 315° to radians.

a. $\frac{7\pi}{5}$

b. $\frac{7\pi}{2}$

c. 2π

d. $\frac{7\pi}{4}$ ✓

$$315 \times \frac{\pi}{180} = \frac{7\pi}{4}.$$

28 Convert 45° to radians.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{5}$

c. $\frac{3\pi}{4}$

d. $\frac{\pi}{2}$

$$45 \times \frac{\pi}{180} = \frac{\pi}{4}.$$

29 Convert 165° to radians.

a. $\frac{11\pi}{6}$

b. $\frac{11\pi}{12}$ ✓

c. π

d. $\frac{11\pi}{13}$

$$165 \times \frac{\pi}{180} = \frac{11\pi}{12}.$$

30 Convert 225° to radians.

a. $\frac{5\pi}{2}$

b. $\frac{5\pi}{4}$ ✓

c. π

d. $\frac{3\pi}{2}$

$$225 \times \frac{\pi}{180} = \frac{5\pi}{4}.$$

31 Convert 150° to radians.

a. $\frac{5\pi}{7}$

b. $\frac{5\pi}{3}$

c. π

d. $\frac{5\pi}{6}$ ✓

$$150 \times \frac{\pi}{180} = \frac{5\pi}{6}.$$

32 Convert 15° to radians.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{13}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{12}$ ✓

$$15 \times \frac{\pi}{180} = \frac{\pi}{12}.$$

33 Convert 75° to radians.

a. $\frac{5\pi}{12}$ ✓

b. $\frac{5\pi}{13}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{2}$

$$75 \times \frac{\pi}{180} = \frac{5\pi}{12}.$$

34 Convert 210° to radians.

a. π

b. $\frac{4\pi}{3}$

c. $\frac{7\pi}{3}$

d. $\frac{7\pi}{6}$ ✓

$$210 \times \frac{\pi}{180} = \frac{7\pi}{6}.$$

35 Convert 240° to radians.

a. $\frac{5\pi}{3}$

b. $\frac{4\pi}{3}$ ✓

c. π

d. $\frac{8\pi}{3}$

$$240 \times \frac{\pi}{180} = \frac{4\pi}{3}.$$

Multiple choice — medium

36 Solve $\tan \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

- a. $120^\circ, 300^\circ$
- b. $45^\circ, 225^\circ$
- c. $30^\circ, 210^\circ$
- d. $135^\circ, 315^\circ$ ✓

$$\tan \theta = -1 \Rightarrow \theta = 135^\circ \text{ or } \theta = 315^\circ.$$

37 Solve $\tan \theta = -\frac{\sqrt{3}}{3}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $30^\circ, 210^\circ$
- b. $150^\circ, 330^\circ$ ✓
- c. $135^\circ, 315^\circ$
- d. $120^\circ, 300^\circ$

$$\tan \theta = -\frac{\sqrt{3}}{3} \Rightarrow \theta = 150^\circ \text{ or } \theta = 330^\circ.$$

38 Solve $\tan \theta = -\sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $150^\circ, 330^\circ$
- b. $120^\circ, 300^\circ$ ✓
- c. $135^\circ, 315^\circ$
- d. $60^\circ, 240^\circ$

$$\tan \theta = -\sqrt{3} \Rightarrow \theta = 120^\circ \text{ or } \theta = 300^\circ.$$

39 Solve $\sin \theta = -\frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $225^\circ, 315^\circ$
- b. $210^\circ, 330^\circ$ ✓
- c. $30^\circ, 150^\circ$
- d. $45^\circ, 135^\circ$

$$\sin \theta = -\frac{1}{2} \Rightarrow \theta = 210^\circ \text{ or } \theta = 330^\circ.$$

40 Solve $\cos \theta = -\frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 330^\circ$

b. $135^\circ, 225^\circ$

c. $150^\circ, 210^\circ$

d. $120^\circ, 240^\circ$ ✓

$$\cos \theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ \text{ or } \theta = 240^\circ.$$

41 Solve $\cos \theta = \frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $150^\circ, 210^\circ$

c. $45^\circ, 315^\circ$

d. $60^\circ, 300^\circ$ ✓

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ \text{ or } \theta = 300^\circ.$$

42 Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $30^\circ, 330^\circ$ ✓

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

$$\cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ \text{ or } \theta = 330^\circ.$$

43 Solve $\cos \theta = \frac{\sqrt{2}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $150^\circ, 210^\circ$

b. $30^\circ, 330^\circ$

c. $45^\circ, 315^\circ$ ✓

d. $120^\circ, 240^\circ$

$$\cos \theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = 45^\circ \text{ or } \theta = 315^\circ.$$

44 Solve $\sin \theta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 150^\circ$

b. $240^\circ, 300^\circ$

c. $60^\circ, 120^\circ$ ✓

d. $210^\circ, 330^\circ$

$$\sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ \text{ or } \theta = 120^\circ.$$

45 Solve $\tan \theta = \sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $60^\circ, 240^\circ$ ✓

b. $135^\circ, 315^\circ$

c. $150^\circ, 330^\circ$

d. $120^\circ, 300^\circ$

$$\tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ \text{ or } \theta = 240^\circ.$$

46 Solve $\tan \theta = \frac{\sqrt{3}}{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $120^\circ, 300^\circ$

b. $30^\circ, 210^\circ$ ✓

c. $135^\circ, 315^\circ$

d. $60^\circ, 240^\circ$

$$\tan \theta = \frac{\sqrt{3}}{3} \Rightarrow \theta = 30^\circ \text{ or } \theta = 210^\circ.$$

47 Solve $\tan \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $60^\circ, 240^\circ$

b. $120^\circ, 300^\circ$

c. $45^\circ, 225^\circ$ ✓

d. $150^\circ, 330^\circ$

$\tan \theta = 1 \Rightarrow \theta = 45^\circ$ or $\theta = 225^\circ$.

48 Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $210^\circ, 330^\circ$

b. $240^\circ, 300^\circ$ ✓

c. $225^\circ, 315^\circ$

d. $60^\circ, 120^\circ$

$\sin \theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = 240^\circ$ or $\theta = 300^\circ$.

49 Solve $\sin \theta = \frac{\sqrt{2}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $60^\circ, 120^\circ$

b. $30^\circ, 150^\circ$

c. $45^\circ, 135^\circ$ ✓

d. $240^\circ, 300^\circ$

$\sin \theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = 45^\circ$ or $\theta = 135^\circ$.

50 Solve $\sin \theta = -\frac{\sqrt{2}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $210^\circ, 330^\circ$

b. $45^\circ, 135^\circ$

c. $60^\circ, 120^\circ$

d. $225^\circ, 315^\circ$ ✓

$$\sin \theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = 225^\circ \text{ or } \theta = 315^\circ.$$

Multiple choice — hard

51 In triangle ABC , $a = 18$, $A = 80^\circ$, $B = 70^\circ$. Find b , to the nearest hundredth.

a. 17.68

b. 18.18

c. 17.18 ✓

d. 16.68

$$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{18 \sin 70^\circ}{\sin 80^\circ} \approx \frac{16.914}{0.985} \approx 17.18.$$

52 In triangle ABC , $a = 17$, $A = 20^\circ$, $B = 80^\circ$. Find b , to the nearest hundredth.

a. 48.95 ✓

b. 49.95

c. 49.45

d. 48.45

$$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{17 \sin 80^\circ}{\sin 20^\circ} \approx \frac{16.742}{0.342} \approx 48.95.$$

53 In triangle ABC , $a = 15$, $A = 60^\circ$, $B = 30^\circ$. Find b , to the nearest hundredth.

a. 9.16

b. 8.66 ✓

c. 8.16

d. 9.66

$$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{15 \sin 30^\circ}{\sin 60^\circ} \approx \frac{7.500}{0.866} \approx 8.66.$$

54 In triangle ABC , $a = 11$, $A = 55^\circ$, $B = 60^\circ$. Find b , to the nearest hundredth.

a. 11.13

b. 12.13

c. 12.63

d. 11.63 ✓

$$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{11 \sin 60^\circ}{\sin 55^\circ} \approx \frac{9.526}{0.819} \approx 11.63.$$

55 In triangle ABC , $a = 17$, $A = 80^\circ$, $B = 45^\circ$. Find b , to the nearest hundredth.

a. 12.71

b. 12.21 ✓

c. 13.21

d. 11.71

$$\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{17 \sin 45^\circ}{\sin 80^\circ} \approx \frac{12.021}{0.985} \approx 12.21.$$

- 56 In triangle ABC , $a = 9$, $b = 15$, and $C = 110^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.

- a. 18.96, 61.4
- b. 19.96, 65.4
- c. 20.96, 63.4
- d. 19.96, 63.4 ✓

$$c^2 = 9^2 + 15^2 - 2(9)(15)\cos 110^\circ \approx 306 + 92.35 \approx 398.35, \text{ so } c \approx 19.96. \text{ Area} \\ = \frac{1}{2}(9)(15)\sin 110^\circ = 135\sin 110^\circ \approx 63.4.$$

- 57 In triangle ABC , $a = 13$, $b = 11$, and $C = 130^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.

- a. 21.77, 56.8
- b. 21.77, 54.8 ✓
- c. 20.77, 52.8
- d. 22.77, 54.8

$$c^2 = 13^2 + 11^2 - 2(13)(11)\cos 130^\circ \approx 290 + 183.84 \approx 473.84, \text{ so } c \approx 21.77. \text{ Area} \\ = \frac{1}{2}(13)(11)\sin 130^\circ = 143\sin 130^\circ \approx 54.8.$$

- 58 In triangle ABC , $a = 13$, $b = 9$, and $C = 80^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.

- a. 14.47, 57.6 ✓
- b. 15.47, 57.6
- c. 14.47, 59.6
- d. 13.47, 55.6

$$c^2 = 13^2 + 9^2 - 2(13)(9)\cos 80^\circ \approx 250 - 40.63 \approx 209.37, \text{ so } c \approx 14.47. \text{ Area} \\ = \frac{1}{2}(13)(9)\sin 80^\circ = 117\sin 80^\circ \approx 57.6.$$

- 59 In triangle ABC , $a = 14$, $b = 10$, and $C = 60^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.

a. 12.49, 60.6 ✓

b. 13.49, 60.6

c. 12.49, 62.6

d. 11.49, 58.6

$$c^2 = 14^2 + 10^2 - 2(14)(10)\cos 60^\circ \approx 296 - 140.00 \approx 156.00, \text{ so } c \approx 12.49. \text{ Area} \\ = \frac{1}{2}(14)(10)\sin 60^\circ = 140\sin 60^\circ \approx 60.6.$$

- 60 In triangle ABC , $a = 9$, $b = 12$, and $C = 70^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.

a. 12.29, 52.7

b. 11.29, 48.7

c. 13.29, 50.7

d. 12.29, 50.7 ✓

$$c^2 = 9^2 + 12^2 - 2(9)(12)\cos 70^\circ \approx 225 - 73.88 \approx 151.12, \text{ so } c \approx 12.29. \text{ Area} \\ = \frac{1}{2}(9)(12)\sin 70^\circ = 108\sin 70^\circ \approx 50.7.$$

- 61 Given $\cos \theta = \frac{35}{37}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $\frac{840}{1369}, \frac{1081}{1369}$ ✓

b. $\frac{840}{1369}, -\frac{1081}{1369}$

c. $-\frac{840}{1369}, \frac{1081}{1369}$

d. $\frac{1081}{1369}, \frac{840}{1369}$

$$\sin \theta = \sqrt{1 - \frac{1225}{1369}} = \frac{12}{37}. \sin 2\theta = 2\left(\frac{12}{37}\right)\left(\frac{35}{37}\right) = \frac{840}{1369}. \cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{288}{1369} = \frac{1081}{1369}.$$

- 62 Given $\cos \theta = \frac{24}{25}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{336}{625}, \frac{527}{625}$

b. $\frac{336}{625}, \frac{527}{625}$ ✓

c. $\frac{336}{625}, -\frac{527}{625}$

d. $\frac{527}{625}, \frac{336}{625}$

$$\sin \theta = \sqrt{1 - \frac{576}{625}} = \frac{7}{25}. \sin 2\theta = 2\left(\frac{7}{25}\right)\left(\frac{24}{25}\right) = \frac{336}{625}. \cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{98}{625} = \frac{527}{625}.$$

- 63 Given $\cos \theta = \frac{15}{17}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $\frac{240}{289}, -\frac{161}{289}$

b. $-\frac{240}{289}, \frac{161}{289}$

c. $\frac{240}{289}, \frac{161}{289}$ ✓

d. $\frac{161}{289}, \frac{240}{289}$

$$\sin \theta = \sqrt{1 - \frac{225}{289}} = \frac{8}{17}. \sin 2\theta = 2\left(\frac{8}{17}\right)\left(\frac{15}{17}\right) = \frac{240}{289}. \cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{128}{289} = \frac{161}{289}.$$

- 64 Given $\cos \theta = \frac{12}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{120}{169}, \frac{119}{169}$

b. $\frac{120}{169}, -\frac{119}{169}$

c. $\frac{119}{169}, \frac{120}{169}$

d. $\frac{120}{169}, \frac{119}{169}$ ✓

$$\sin \theta = \sqrt{1 - \frac{144}{169}} = \frac{5}{13}. \sin 2\theta = 2\left(\frac{5}{13}\right)\left(\frac{12}{13}\right) = \frac{120}{169}. \cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{50}{169} = \frac{119}{169}.$$

- 65 Given $\cos \theta = \frac{4}{5}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{24}{25}, \frac{7}{25}$

b. $\frac{24}{25}, \frac{7}{25}$ ✓

c. $\frac{24}{25}, -\frac{7}{25}$

d. $\frac{7}{25}, \frac{24}{25}$

$$\sin \theta = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}. \sin 2\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}. \cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{18}{25} = \frac{7}{25}.$$